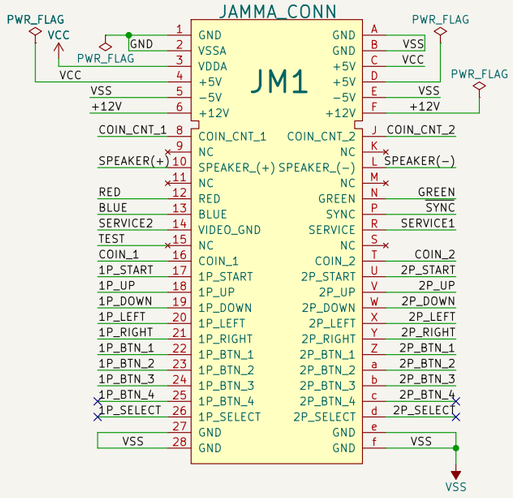
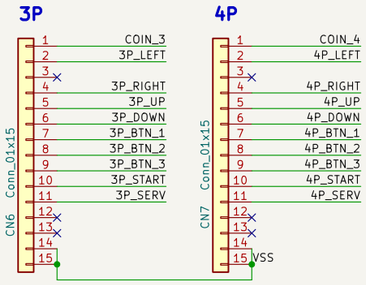
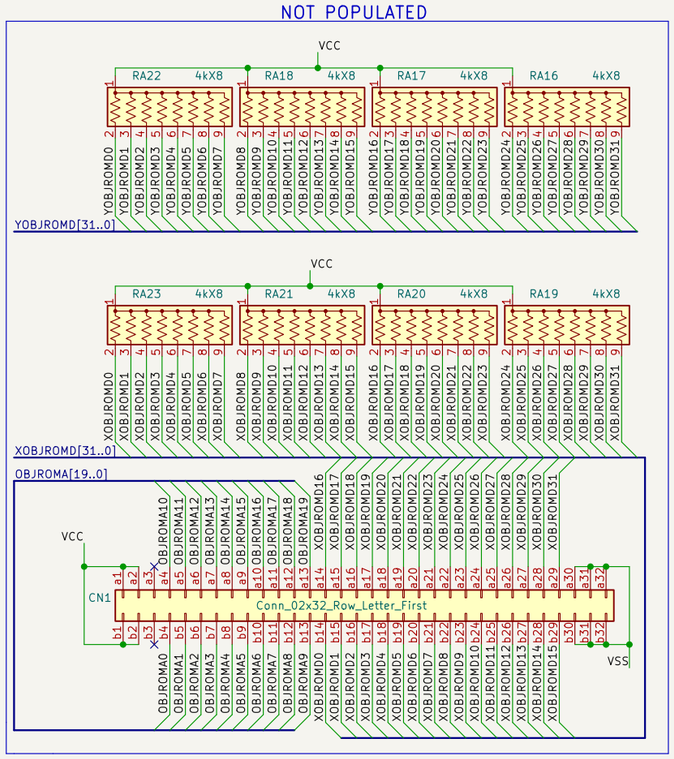
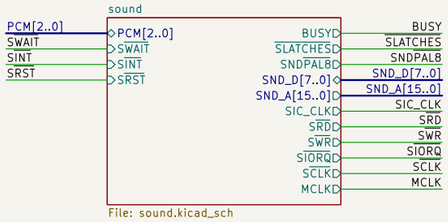
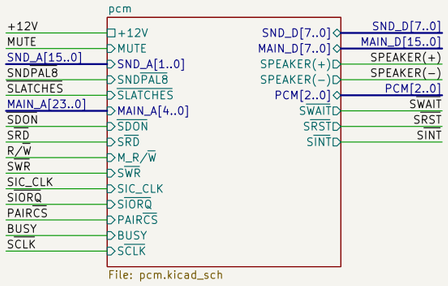
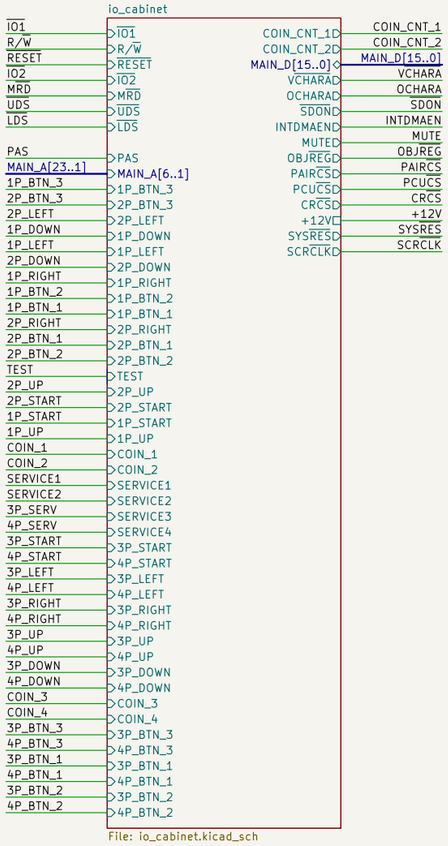


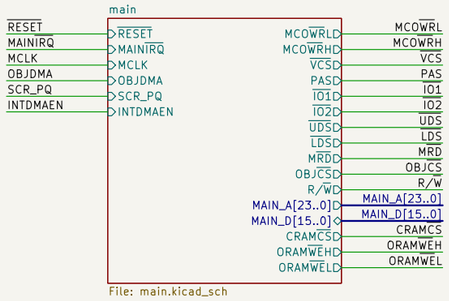


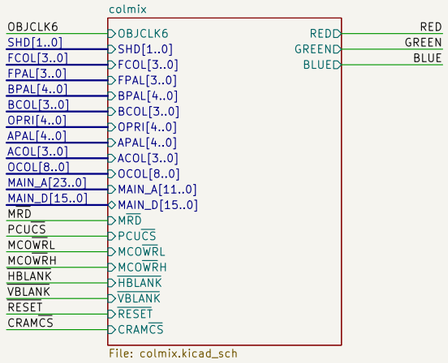




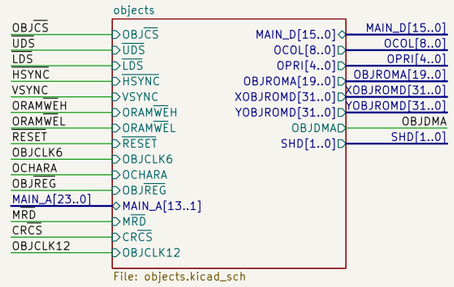


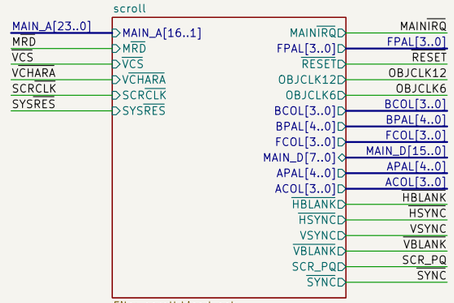










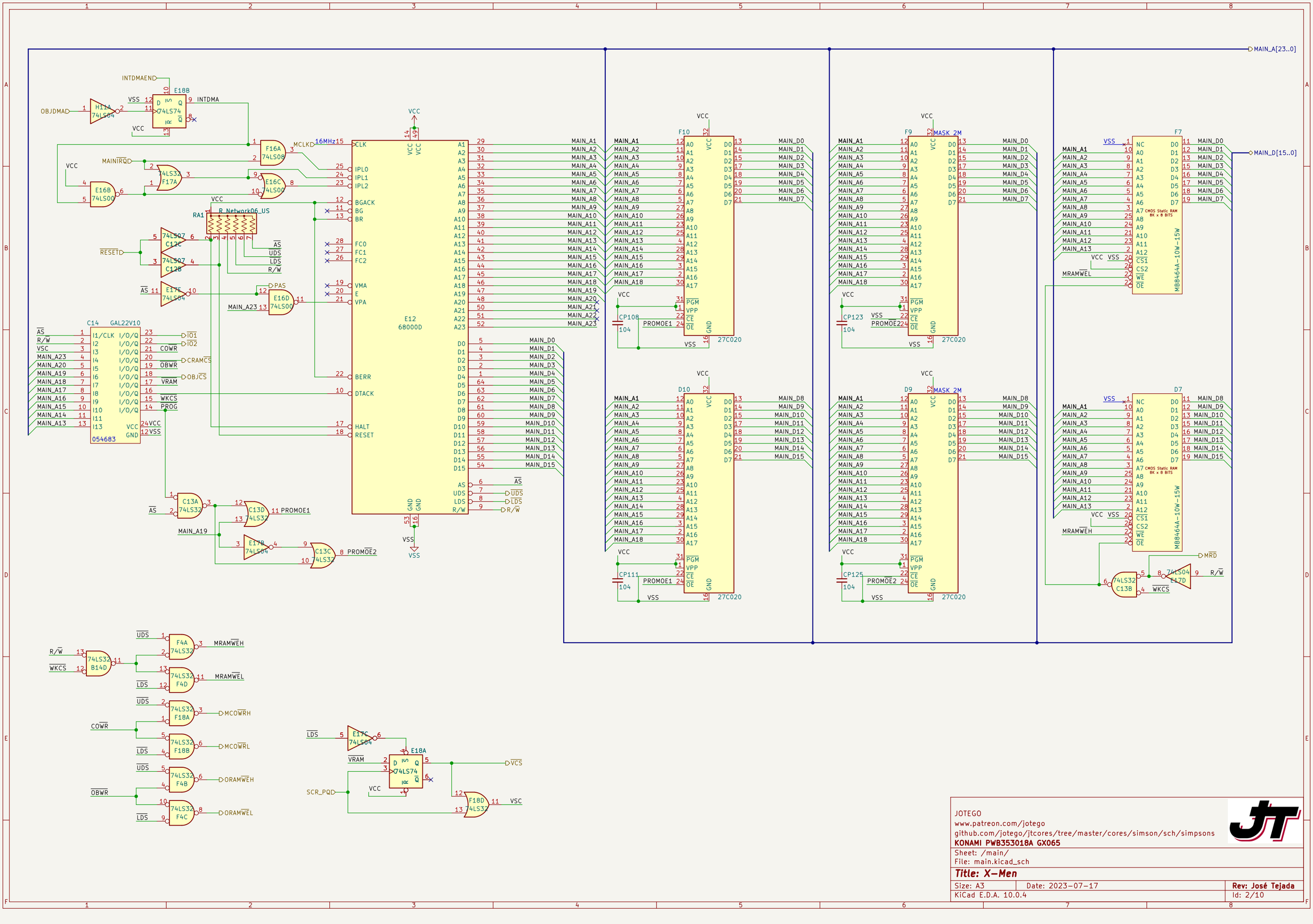


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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /
File: xmen.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 1/10



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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /main/
File: main.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 2/10

S34
3P_BTN_3
4P_BTN_3
COIN_3
COIN_4
R3P_BTN_3
R4P_BTN_3
RCOIN_3
RCOIN_4
KONAMI_005273
S33
3P_BTN_1
4P_BTN_1
3P_BTN_2
4P_BTN_2
R3P_BTN_1
R4P_BTN_1
R3P_BTN_2
R4P_BTN_2
KONAMI_005273
S37
3P_LEFT
4P_LEFT
3P_RIGHT
4P_RIGHT
R3P_LEFT
R4P_LEFT
R3P_RIGHT
R4P_RIGHT
KONAMI_005273
S36
3P_UP
4P_UP
3P_DOWN
4P_DOWN
R3P_UP
R4P_UP
R3P_DOWN
R4P_DOWN
KONAMI_005273
S40
1P_BTN_3
2P_BTN_3
COIN_1
COIN_2
R1P_BTN_3
R2P_BTN_3
RCOIN_1
RCOIN_2
KONAMI_005273
S39
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
R1P_BTN_1
R2P_BTN_1
R1P_BTN_2
R2P_BTN_2
KONAMI_005273
S43
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
R1P_LEFT
R2P_LEFT
R1P_RIGHT
R2P_RIGHT
KONAMI_005273
S42
1P_UP
2P_UP
1P_DOWN
2P_DOWN
R1P_UP
R2P_UP
R1P_DOWN
R2P_DOWN
KONAMI_005273
C17
R3P_BTN_1
R4P_BTN_1
R3P_BTN_2
R4P_BTN_2
RCOIN_3
RCOIN_4
R3P_BTN_3
R4P_BTN_3
MAIN_A1
JOYSTICK
MAIN_D12
MAIN_D13
MAIN_D15
MAIN_D14
74LS257
C18
R3P_UP
R4P_UP
R3P_DOWN
R4P_DOWN
R3P_RIGHT
R4P_RIGHT
R3P_LEFT
R4P_LEFT
MAIN_D8
MAIN_D9
MAIN_D11
MAIN_D10
74LS257
C15
R1P_BTN_1
R2P_BTN_1
R1P_BTN_2
R2P_BTN_2
RCOIN_1
RCOIN_2
R1P_BTN_3
R2P_BTN_3
MAIN_D4
MAIN_D5
MAIN_D7
MAIN_D6
74LS257
CC15
R1P_UP
R2P_UP
R1P_DOWN
R2P_DOWN
R1P_RIGHT
R2P_RIGHT
R1P_LEFT
R2P_LEFT
MAIN_D0
MAIN_D1
MAIN_D3
MAIN_D2
74LS257
MAIN_D[15..0]
TEST
SW17
EVQ-QS5-07K
switch
button
R13
220Ω
R14
3.3KΩ
VCC
CAB14
CC3
104
VSS
CAB[15..0]
S44
1P_START
2P_START
3P_START
4P_START
CAB8
CAB9
CAB10
CAB11
KONAMI_005273
S54
SERVICE1
SERVICE2
SERVICE3
SERVICE4
CAB0
CAB1
CAB2
CAB3
KONAMI_005273
C16
CAB8
CAB9
CAB10
CAB11
CAB14
OTHER
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
MAIN_D14
MAIN_D15
3-StateHexBuffers
74LS367
VCC
R33
4.7KΩ
SW-B12
OFF
ON
SLIDESW1P
NOT
MOUNTED
VSS
D16
CAB0
CAB1
CAB2
CAB3
CAB6
CAB7
OTHER
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D6
MAIN_D7
3-StateHexBuffers
74LS367
D17
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
OCHARA
VCHARA
SDON
MUTE
RESET
FOUR FLIP FLOP WITH
DOUBLE RAIL OUTPUTS
74LS175
UDS
74LS32
F17D
CC13
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D4
MAIN_D5
INTDMAEN
74LS174
hex flip flop
13C label is
duplicated on
the PCB
B13
ER5911
EEPROM
128X8 or 64X16
CAB6
CAB7
VCC
FL-B12
+12V
VSS
MAIN_A1
MAIN_A6
MAIN_A5
IO1
MAIN_A3
MAIN_A2
MAIN_A[6..1]
MRD
IO2
R/W
E17A
74LS04
A1 D15B O0
A0 O1
O2
E 74LS139 O3
decoder
PAIRCS
PCUCS
OBJREG
LDS
74LS32
B14B
B14C
74LS32
B14A
74LS32
JOYSTICK
OTHER
CRCS
F5B
74LS74
PAS
AFE
VCC
VSS
AFE
JP-A12
SJO
CP76
C_Small
VSS
A11
CLK
VREF
RES
CT
VS
OUT1
OUT2
+5V
+12V
RESET
CNT2
CNT2
CNT1
CNT1
N.C
Master/RESET generation
for the whole system
and watchdog timer
KONAMI
051550
COIN_CNT_2
COIN_CNT_1
G18A
74LS74
SYSRES
VCC
G12
24MHz
XTAL_OSC
VSS
H11C
74LS04
SCRCLK
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch
Title: X-Men
Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 3/10

NOT POPULATED

VCC

RA24 4.7kX8 RA25 4.7kX8 RA26 4.7kX8 RA27 4.7kX8

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7 GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15 GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23 GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

DIN64SS
CNS
Connector First

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7 GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15 GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23 GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMA[18..0]

J17
H17
74LS373
LATCHES AND FLIP FLOPS

GFXROMA0 GFXROMA1 GFXROMA2 GFXROMA3 GFXROMA4 GFXROMA5 GFXROMA6 GFXROMA7 GFXROMA8 GFXROMA9 GFXROMA10 GFXROMA11 GFXROMA12 GFXROMA13 GFXROMA14 GFXROMA15 GFXROMA16 GFXROMA17 GFXROMA18

GFXROMA'0 GFXROMA'1 GFXROMA'2 GFXROMA'3 GFXROMA'4 GFXROMA'5 GFXROMA'6 GFXROMA'7 GFXROMA'8 GFXROMA'9 GFXROMA'10 GFXROMA'11 GFXROMA'12 GFXROMA'13 GFXROMA'14 GFXROMA'15 GFXROMA'16 GFXROMA'17 GFXROMA'18

GFXROMA[18..0]

L15 L16
065A08 KONAMI
MASK 8M – 16BIT
CMOS ROM
054903

GFXROMA'[18..0]

GFXROMD[31..0]

APAL[4..0] ACOL[3..0] SCRA FIX FCOL[3..0] FPAL[3..0]

H11D 74LS04 VSYNC

J14
KONAMI
051962
TILE DRAWING

LAYER A TRANSPARENCY
LAYER A COLOR INDEX
GFX ROM DATA
FIX LAYER TRANSPARENCY
FIX LAYER COLOR INDEX
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
ATTRIBUTE CODE FROM 052109
CPU ADDRESS
MAIN CLK
LAYER A H SCROLL
LAYER B H SCROLL
READ GFX ROM ENABLE
SOUND

MAIN_D[7..0]
BCOL[3..0] SCRB
BPAL[4..0]
OBJCLK6 OBJCLK12
MAIN_D0 MAIN_D1 MAIN_D2 MAIN_D3 MAIN_D4 MAIN_D5 MAIN_D6 MAIN_D7
BCOL0 BCOL1 BCOL2 BCOL3
BPAL0 BPAL1 BPAL2 BPAL3 BPAL4

OBJCLK12
H11B 74LS04
74LS74

SCRA[12..0]

F13 F14 F15
RAM0 RAM1 RAM2
CMOS Static RAM 8K x 8 BITS
MB8464A-80-10-15/L-LL

VD[15..0]

D18
74LS367
3-StateHexBuffers
R7 470Ω
R5 100Ω
CSYNC SYNC

G14
KONAMI
052109
TILE MAP GENERATOR
30 pins per side

TILE ATTRIBUTES
GFX ROM ADDRESS
LAYER A SCROLL X
LAYER A SCROLL X ZA1H
LAYER B SCROLL X ZB3H
MAIN CLK
CPU ADDRESS
GFX ROM read
VRAM ADDRESS
VRAM DATA
CPU DATA

MAIN_A[16..1]

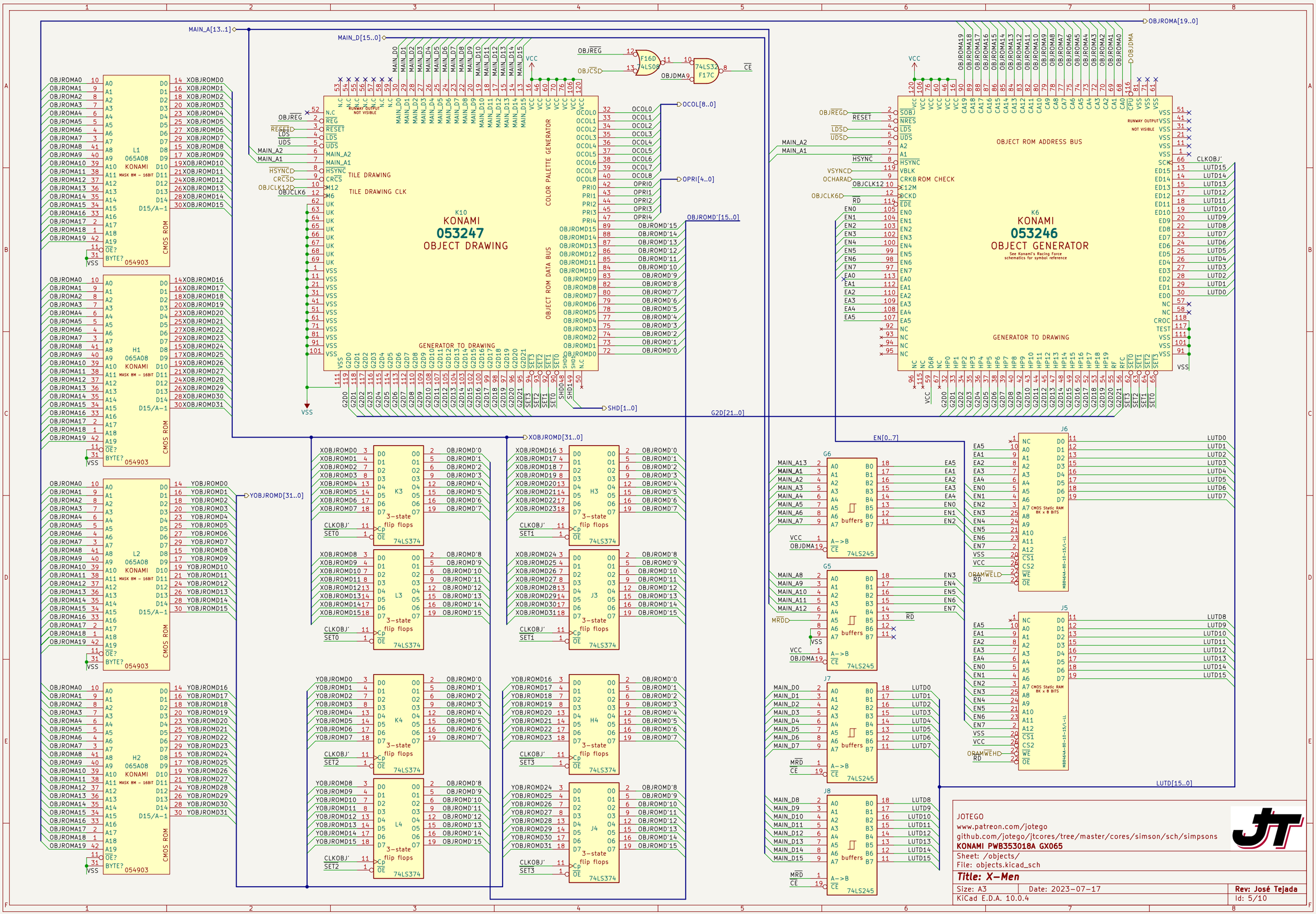
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons

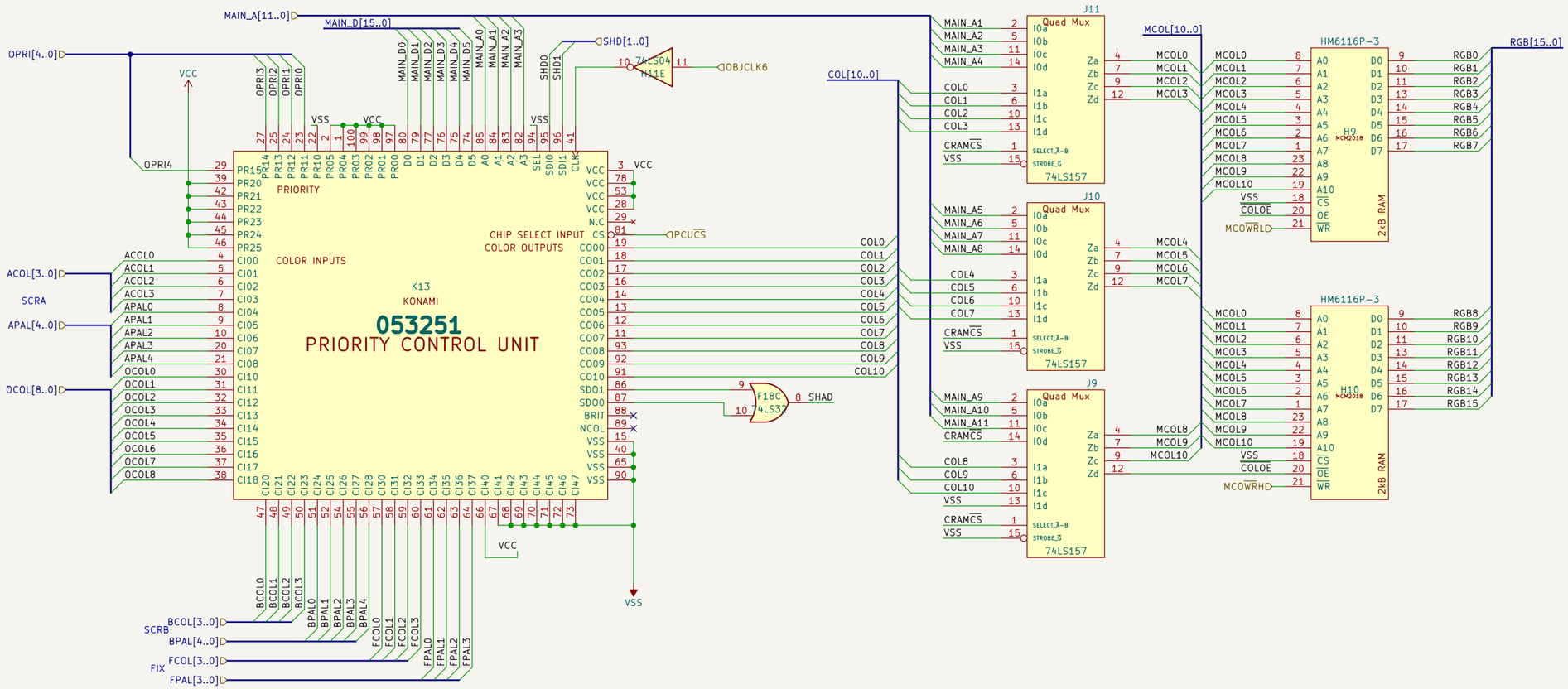
KONAMI PWB353018A GX065

Sheet: /scroll/
File: scroll.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	Rev: José Tejada
KiCad E.D.A. 10.0.4		Id: 4/10


MAIN_A[13..1]
MAIN_D[15..0]
OBJROMA[19..0]
OBJREG
OBJCS
F16D
74LS08
74LS32
F17C
OBJDMA9
CE
VCC
K10
KONAMI
053247
OBJECT DRAWING
TILE DRAWING
TILE DRAWING CLK
COLOR PALETTE GENERATOR
OBJECT ROM DATA BUS
GENERATOR TO DRAWING
OCOL[8..0]
OPRI[4..0]
OBJROMD'[15..0]
SHD[1..0]
G2D[21..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
K6
KONAMI
053246
OBJECT GENERATOR
See Konami's Racing Force schematics for symbol reference
OBJECT ROM ADDRESS BUS
ROM CHECK
GENERATOR TO DRAWING
CLKOBJ'
LUTD15
LUTD14
LUTD13
LUTD12
LUTD11
LUTD10
LUTD9
LUTD8
LUTD7
LUTD6
LUTD5
LUTD4
LUTD3
LUTD2
LUTD1
LUTD0
LUTD[15..0]
EN[0..7]
L1
L2
H1
H2
K3
K4
L3
L4
H3
H4
J3
J4
J5
J6
G5
G6
J7
J8
065A08
MASK 8M - 16BIT
054903
CMOS ROM
3-state flip flops
74LS374
buffers
74LS245
CMOS Static RAM 8K x 8 BITS
M88P64A-80-10-15/L-LL
JOTEGO
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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065
Sheet: /objects/
File: objects.kicad_sch
Title: X-Men
Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 5/10

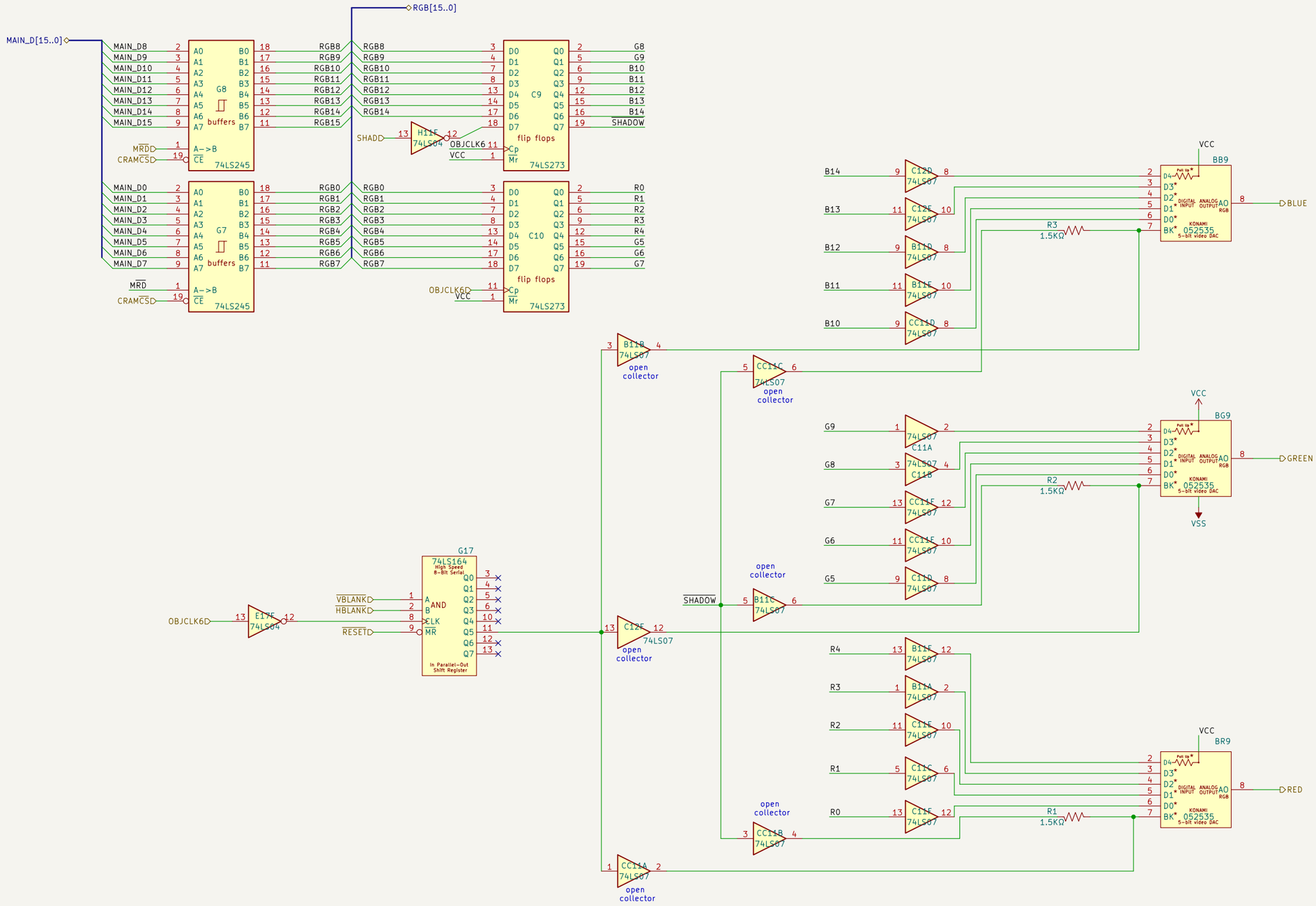


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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /colmix/
File: colmix.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 6/10

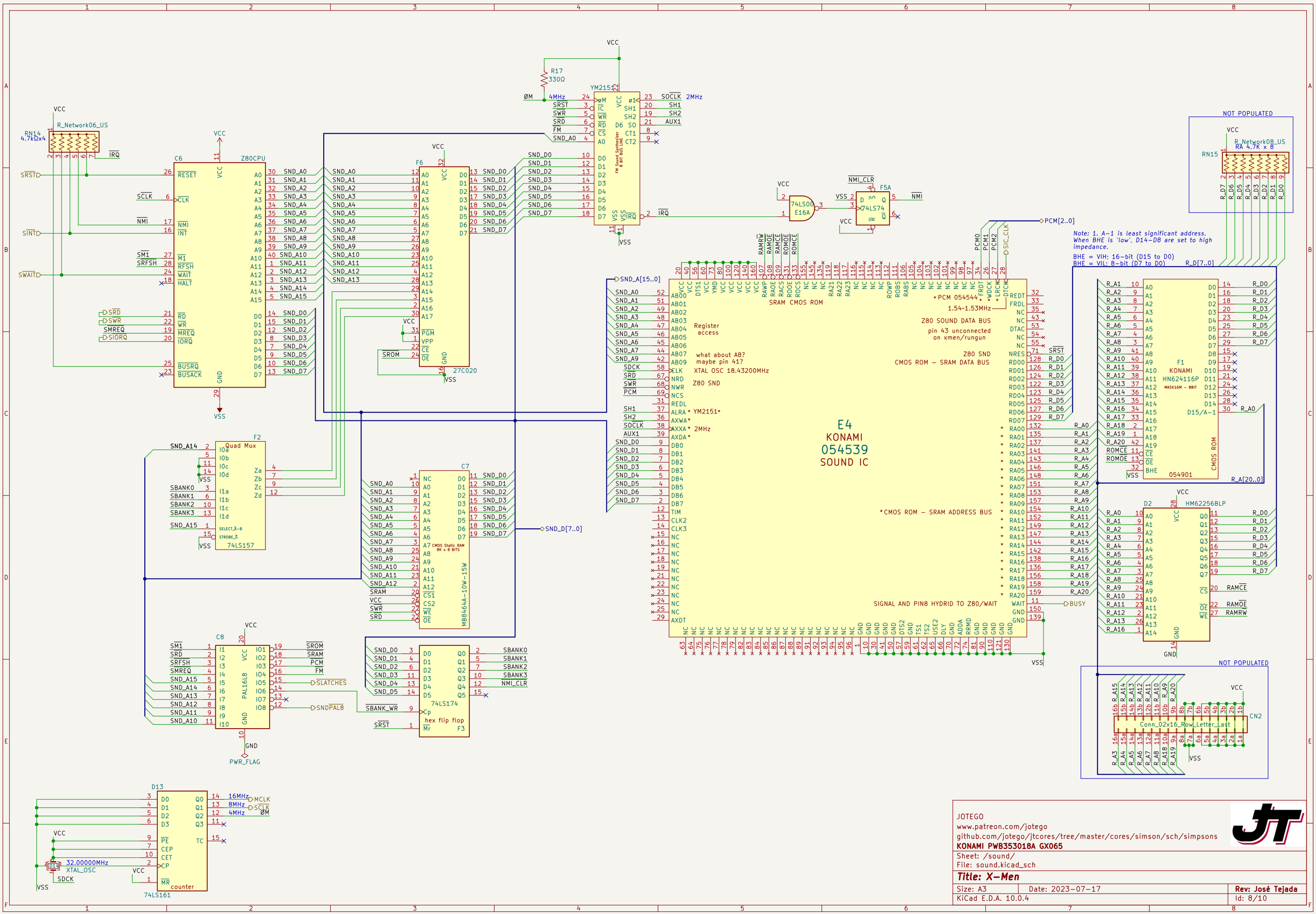


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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 7/10



Note: 1. A–1 is least significant address.
When BHE is 'low', D14–D8 are set to high impedance.

BHE = VIH; 16-bit (D15 to D0)
BHE = VIL; 8-bit (D7 to D0)

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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons

KONAMI PWB353018A GX065

Sheet: /sound/
File: sound.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 8/10

KONAMI 054544

Audio module

Sound communication latch/volume manager chip

THIS CONNECTOR REPLACES THE PINS OF THE KONAMI CI 054544

C4-1 KONAMI 054321 PCM

054539 SND IC

C4-2 AD1868R Triple Supply Out 18-Bit Audio DAC

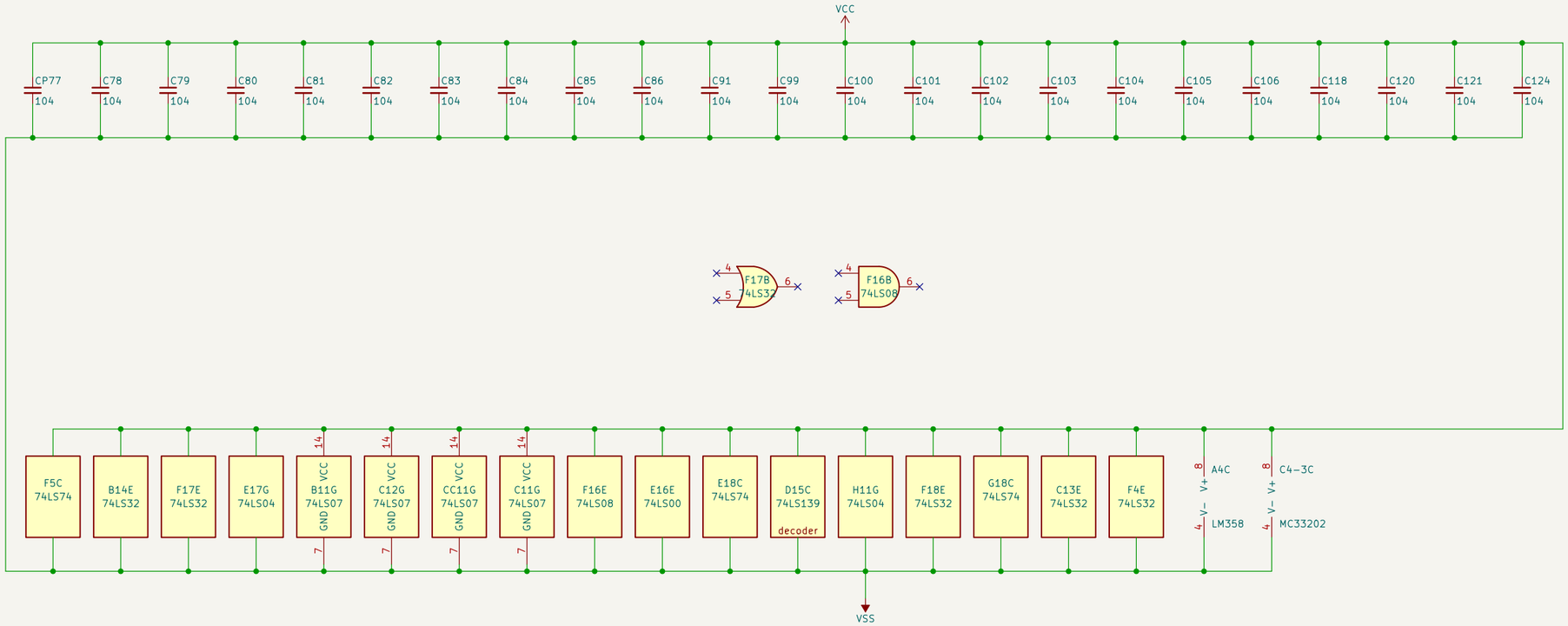
Pins 8 and 16 of the AD1868R are connected to pins 40 and 36 of the 054544 respectively, but these pins (40 and 36) are floating.

SHORTED ON MOTHER PCB

SOUND AMPLIFIER MOTHER PCB

JOTEGO	
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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons	
KONAMI PWB353018A GX065	
Sheet: /pcm/	
File: pcm.kicad_sch	
Title: X-Men	
Size: A3	Date: 2023-07-17
KiCad E.D.A. 10.0.4	Id: 8/10

Rev: José Tejada



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github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 9/10